

Descriptive Statistics

Figure 1 below shows that nearly 30% of the telecommunications customers have churned. This is an alarmingly high figure given the fierce competition in the telecommunication industry and the challenges in attracting and retaining customers. The result indicates that the company is not targeting the correct audiences or not implementing the most effective initiatives and strategies to retain their customers.

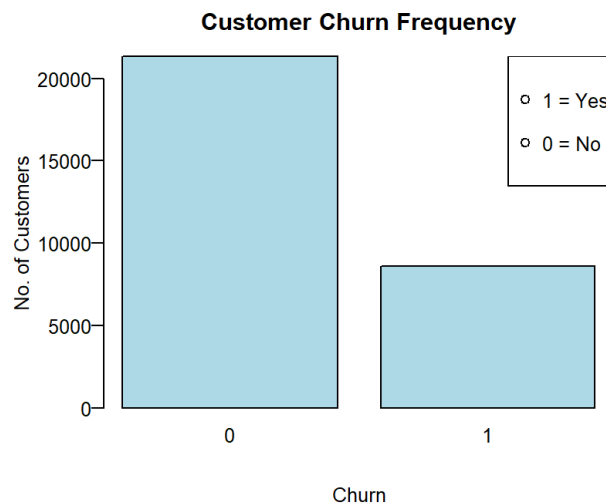


Figure 1. Numbers of Churned and Not Churned Customers

The summary of monthly revenue for both churned and not churned customers is shown in Table 1. Customers who choose to stay with the company generated more revenue per month for the company than customers who decided to use another mobile service provider. The average revenue that not churned customers is \$77.86, relatively higher than \$69.84 of churned customers. In addition, churned customers have a much lower maximum monthly revenue of only \$861, while that of not churned customers is over \$1200. Hence, it is critical for the telecommunication company to reduce the number of churned customers to maximise their profit and stay competitive.

	Churned customers	Unchurned customers
Minimum	0	-5.86
Mean / Average	69.84	77.86
Maximum	861.11	1223.38
1 st quartile	37.37	40.05
3 rd quartile	85.44	97.81

Table 1. Monthly Revenue of Churned and Unchurned Customers

Analysing the age distribution of the customer household members can also provide helpful insights into the customer demographics. The average age of the first household members was found to be 31.33 years and the average age for the second household members was 21.16 years. The results indicate that there is a significant difference in generation between the two groups. Furthermore, the analysis found that churned customers appear to have a younger first member of the household and older-first household members is more likely to stay with the company.

Customers' income tends to have the largest impact on their spending behaviour and whether they choose to move to another company. As can be seen from Figure 2, income group 0 has the largest number of churned customers with the assumption of customers this group has the lowest income. This suggests that lower income customers tend to have higher churn rate than high-income earners. One interesting finding is that group 6 has the second highest number of churned customers just below group 0. Based on the assumption, people belonging to group 6 are medium-income earners with stable financial and potentially better credit scores.

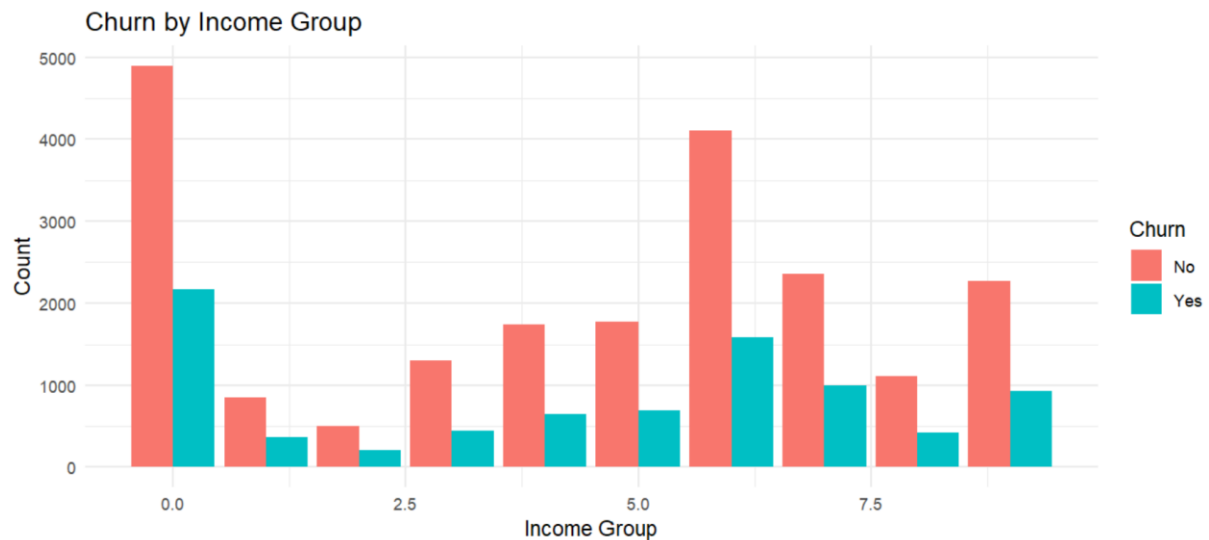


Figure 2. Income Groups of Churned and Unchurned Customers

Another important factor that can impact customers' decisions to churn is their credit ratings. The pie chart in Figure 3 compares the credit ratings between churned and unchurned customers. The results are similar for both churned and unchurned customers. Churned customers with higher credit score imply that customers with financial stability and creditworthiness probably have a wider choice of service providers. They may also be more responsible in managing their finances, which makes them more active in search of finding better offers or better services. Therefore, they are less likely to stay with a provider that they deem is overvalued for their money. The results are aligned with the above observations related to the customer income, suggesting that this group of customers may be more value conscious and difficult to attract.

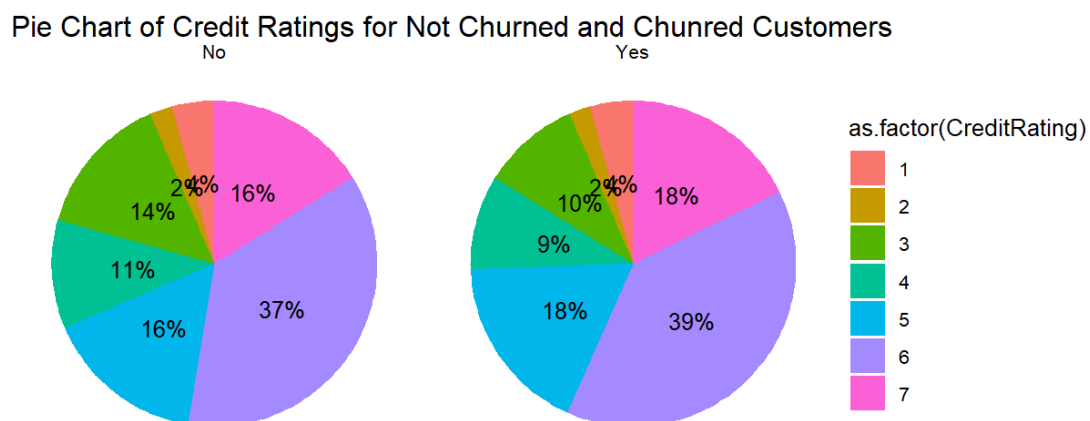


Figure 3. Credit Ratings for Unchurned and Churned Customers

Statistical Modelling

Before building any statistical model, the given data set of 30,000 customers is split into training data set and testing data set to ensure an unbiased evaluation of all models. The training set consists of 70% of the data, and the testing set accounts for the remaining 30%. The training data set will be used to construct a random forest decision tree, a k-nearest neighbor (KNN) and a Naïve Bayesian (NB) model, while the testing set will be used to evaluate the performance of these models. For more details about the three models, please refer to the Appendix.

Model selection

Table 2 shows the performance measures for all three models with the cutoff equals to 0.5. The random forest decision tree model and the NB model have quite similar accuracy of 71.1% and 71.4% respectively, while the KNN model's accuracy is only 66.8%. All three models have significantly low sensitivity with the highest equals to 3.1% for random forest model. This means that these models do a poor job in correctly classifying target class cases, which is identifying churned customers.

On the other hand, the specificity of these models is exceptionally high which indicates the models are better at identifying unchurned customers. The ROC curve shows the sensitivity and specificity measures across all cutoff values, and the higher the area under the curve (AUC) the better. The random forest model has an AUC of 52% which is the highest among three models. Overall, the random forest model performs slightly between the KNN and NB model.

	Random forest	KNN	NB
Accuracy	71.1%	66.8%	71.4%
Sensitivity	3.1%	1.7%	1.7%
Specificity	98.6%	89.0%	99.0%
ROC / AUC	52.0%	49.7%	50.8%

Table 2. Performance Measures of Random Forest, KNN and NB Model

However, the given data set faced a huge imbalance issue with 71% of its data belongs to the non-target class of unchurned customers and only 29% is target class of churned customers. Imbalanced data is problematic since it can cause the models to be biased toward the majority class, which in this is the non-target class of unchurned customers. The accuracy scores can also be a misleading indicator of the model quality. To address this problem, three more random forest models are constructed using the cutoff equals to prevalence, random undersealing the majority class and random oversampling minority class technique respectively. The performance measures for these three models together with the original random forest model are shown in Table 3 below.

	Method	Accuracy	Sensitivity	Specificity	PPV	NPV
3	Random Undersampling Majority Class	0.568	0.496	0.598	0.332	0.746
2	Raw Imbalanced Data: Cutoff = Prevalence	0.61	0.349	0.716	0.332	0.731
4	Random Oversampling Minority Class	0.665	0.224	0.843	0.365	0.729
1	Raw Imbalanced Data: Cutoff = 0.5	0.71	0.044	0.98	0.465	0.717

Table 3. Performance measures of Random Forest Models

The original model with imbalanced data and cutoff equals to 0.5 has the highest accuracy and specific rate of 71% and 98% respectively, while random under sampling majority class model has the highest sensitivity equals to 49.6%. Given the focus of correctly identifying the target class of churned customers in this case, the random under sampling model appears to be the most appropriate model. In addition, the OOB plots (Appendix F) show the out of bag error for each random forest tree to measure how good the model is going on unseen data. With the most stabilised OOB plot meaning adding more trees would not benefit the analysis, the random under sampling model is selected for further analysis and prediction on unseen data.

Random under sampling majority class model

The final random under sampling forest decision tree is constructed using the target variable Churn, and other predictor variables such as monthly revenue and monthly minutes for the company, the age of first and second household member, retention calls, income group, credit ratings and occupation of the customers. These predictor variables are chosen based on the previous descriptive analysis that they are important factors and can impact customer's decision to churn.

To further analyse the performance of the selected model, a cumulative chart lift is generated to determine at the point which the model's predictions become less useful. As the lift curve in black is above the base line in red (Figure 4), this is an indicative of a good performance of the selected model.

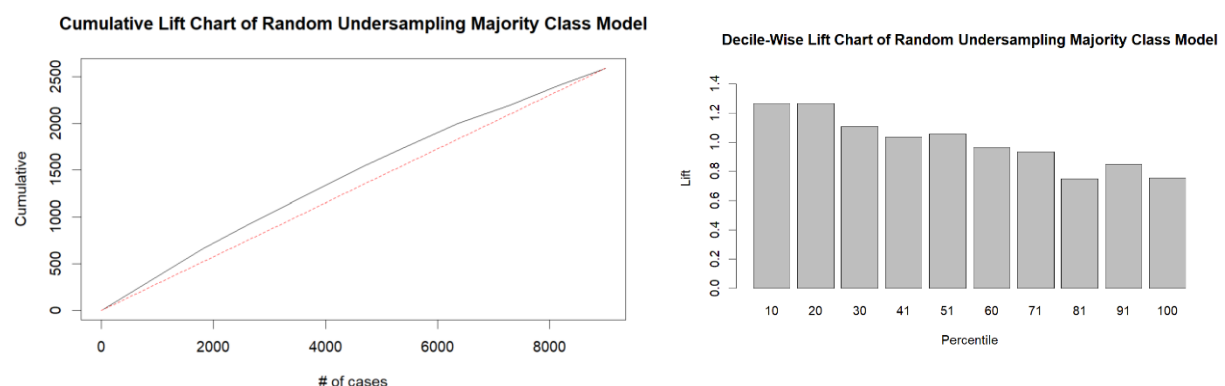
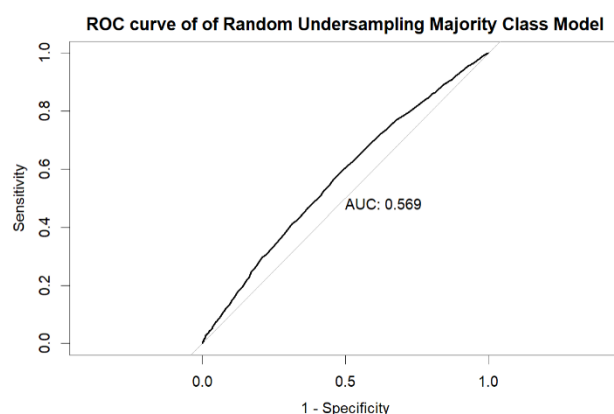


Figure 4. Cumulative chart & Decile-Wise Lift chart of Selected model

Similar information is conveyed through the Decile-wise life chart (Figure 5) but in percentiles instead. It is crucial to note that the lift of the percentiles decreases gradually over time and eventually falls below 1.0. This suggests that there is still room for improvement since the model seemed to have moderate performance.

The ROC curve is examined to better understand the sensitivity and specificity measures of both models across all cutoff values. The ROC curve lies above the baseline suggests that the selected model has a moderate predictive performance with the AUC of 56.9%. In conclusion, the random under sampling of the majority class model has a moderate predicted performance.



Main findings and Recommendations

Based on the above analysis, it seems like customer characteristics such as the numbers they have called the retention team, the age of the household members and their income levels are the three most important factors that determine whether a customer will churn or not. This information is useful to help develop the targeted retention strategies based on these characteristics and behaviour of the customers.

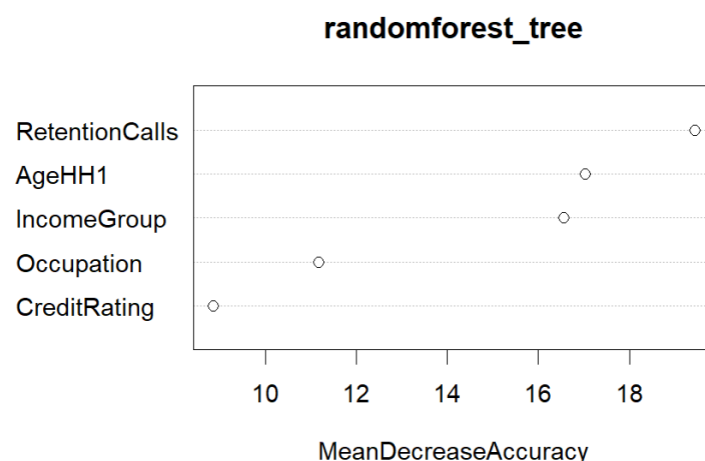
It is recommended that the telecommunication company refocus its primary customer groups based on income level. The above analysis shows that people with low income are the main group of customers in the company, but they are more likely to switch compared to high-income earners and can require more resources to retain. Therefore, the company should focus its effort on attracting more customers with higher incomes since they appear to be easier to retain and more loyal.

Furthermore, the middle-income groups show huge potential in helping the company reach its objectives. They are the second largest group of customers in the company but also more likely to switch to another service providers. This indicates that the company needs to offer more valuable deals to make its services appear cost-effective to value conscious customers.

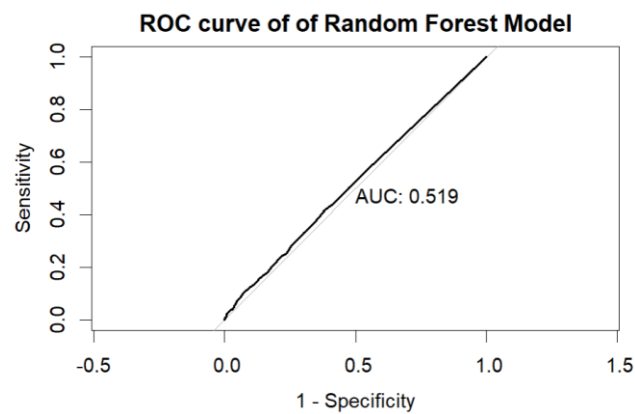
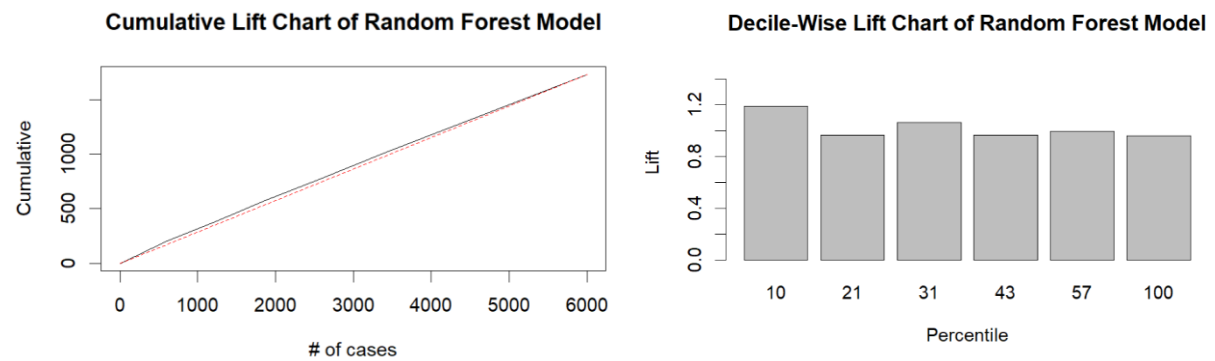
The second suggestion for the company is to launch more targeted marketing campaigns and promotional efforts tailored to the older age group since they appear to be more loyal to the company. In addition, older customers are also less likely to switch from one service provider to another given the familiarity and convenience. The company should take advantage of knowing the average ages of primary and secondary household members to segment its customers and provide more customized products and customer service to the age groups.

Appendix

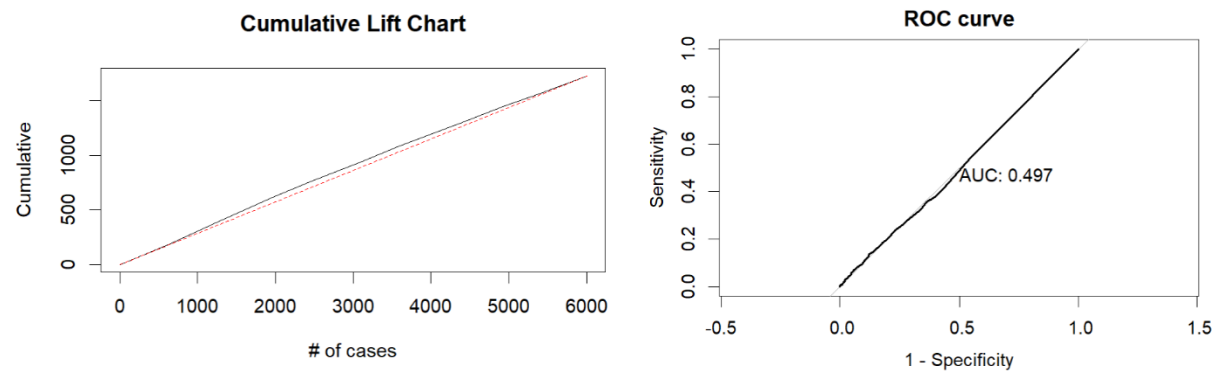
Appendix A. Random Forest Variable Plot



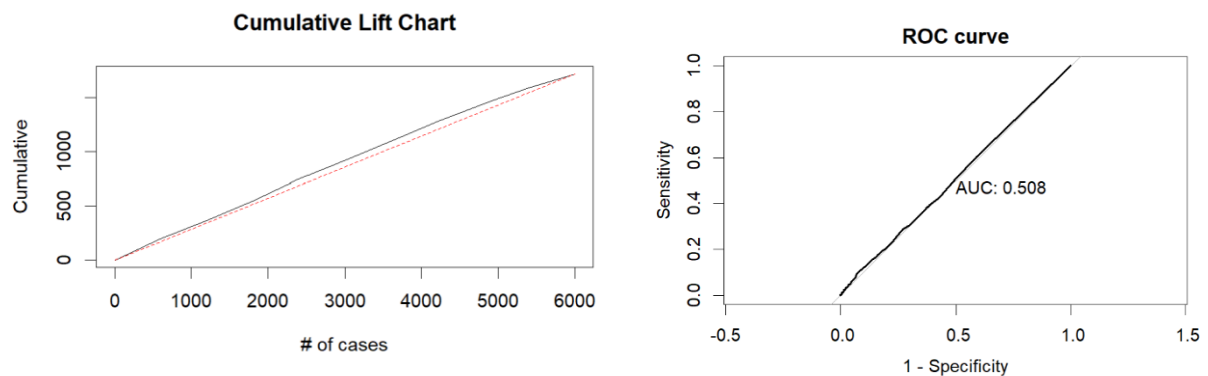
Appendix B. Performance measure charts of Random Forest Model



Appendix C. Performance measure charts of Random Forest Model

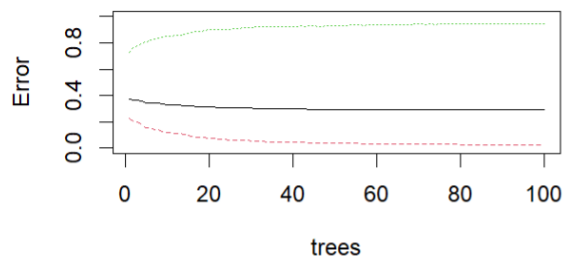


Appendix D. Performance measure charts of Random Forest Model

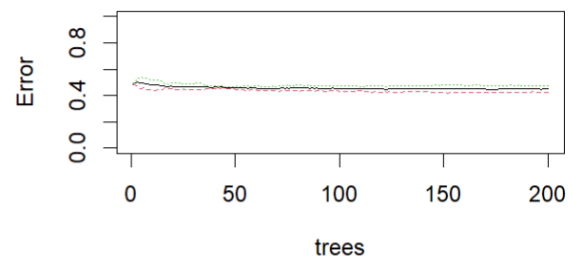


Appendix F. OOB Plots

Raw Imbalanced Data: Cutoff = 0.5



Random Undersampling Majority Class



Random Oversampling Minority Class

